Task 05 - Part 1 of 2: SQL Aggregations & GROUP BY

Course: Database Applications Development
Lesson: 05 - SQL Aggregations, Grouping, and Excel Export (in Part 2)

Instructions

Complete all exercises in this notebook. You will:

1. Write SQL queries using aggregate functions
2. Use GROUP BY to analyze categories
3. Export 4 query results to Excel files
4. Answer analysis questions

Resources:

- Lesson materials (dbApps05_AggregationsGrouping.md)
- Walkthrough notebook (dbApps05_Walkthrough.ipynb)
- SQL Reference Guide (updated with aggregations)

Submission:

1. Complete all TODO sections
2. Verify all cells run without errors
3. Check that Excel files were created (in Part 2)
4. Push to GitHub: `databaseApplications/dbApps05TasksPart1.ipynb`

Let's practice aggregations!

Setup

In [69]:
```
import pandas as pd
import sqlite3

# Connect to database
conn = sqlite3.connect('nba_5seasons.db')
print("✅ Connected to database")
```

```
✅ Connected to database
```

Part 1: Basic Aggregate Functions (10 queries)

Practice using COUNT, SUM, AVG, MIN, and MAX without GROUP BY.

Hints:

- Remember to use AS to name your result columns
- Aggregate functions work on ALL rows that match your WHERE clause
- Use ROUND(AVG(column), 1) to round decimals

Query 1: Count All Teams

Task: How many teams are in the database?

Hint: Use COUNT(*) on the teams table.

In [73]:
```
# TODO: Write your query
query_1 = """
SELECT COUNT(*) AS total_teams
FROM teams
"""

result_1 = pd.read_sql(query_1, conn)
display(result_1)
```

	total_teams
0	30

Query 2: Count Player Season Records

Task: How many player-season records exist for 2021-22?

Hint: COUNT(*) from player_season_stats WHERE season = '2021-22'

In [76]:
```
# TODO: Write your query
query_2 = """
SELECT COUNT(*) AS total_players
FROM player_season_stats
WHERE season = '2021-22'
"""

result_2 = pd.read_sql(query_2, conn)
display(result_2)
```

	total_players
0	605

Query 3: Total Points (All Teams)

Task: What were the total combined points scored by all teams in all 2021-22 games?

Hint: SUM(pts) from team_game_stats WHERE season = '2021-22'

In [79]:
```
# TODO: Write your query
query_3 = """
SELECT SUM(pts) AS total_points
FROM team_game_stats
WHERE season = '2021-22'
"""

result_3 = pd.read_sql(query_3, conn)
display(result_3)
```

	total_points
0	272115

Query 4: Average Points Per Game (League-Wide)

Task: What was the league-wide average points per game in 2021-22?

Hint: AVG(pts), remember to round to 1 decimal place

In [82]:
```
# TODO: Write your query
query_4 = """
SELECT ROUND(AVG(pts), 1) AS avg_pts
FROM team_game_stats
WHERE season = '2021-22'
"""

result_4 = pd.read_sql(query_4, conn)
display(result_4)
```

	avg_pts
0	110.6

Query 5: Highest and Lowest Scores

Task: Find the highest and lowest points scored in any single game during 2021-22.

Hint: Use both MAX(pts) and MIN(pts) in one query

In [85]:
```
# TODO: Write your query
query_5 = """
SELECT
MIN(pts) AS min_points,
```

```
MAX(pts) AS max_points
FROM team_game_stats
WHERE season = '2021-22'
"""

result_5 = pd.read_sql(query_5, conn)
display(result_5)
```

	min_points	max_points
0	75	158

Query 6: Lakers Total Points

Task: How many total points did the Lakers (team_id = 1610612747) score in 2021-22?

Hint: SUM(pts) with WHERE for team_id AND season

In [88]:
```
# TODO: Write your query
query_6 = """
SELECT SUM(pts) AS lakers_total_points
FROM team_game_stats
WHERE team_id = 1610612747 AND season = '2021-22'
"""

result_6 = pd.read_sql(query_6, conn)
display(result_6)
```

	lakers_total_points
0	9192

Query 7: Warriors Average Points

Task: What was the Warriors' (team_id = 1610612744) average points per game in 2021-22?

Hint: AVG(pts), round to 1 decimal

In [91]:
```
# TODO: Write your query
query_7 = """
SELECT ROUND(AVG(pts), 1) AS avg_points
FROM team_game_stats
WHERE team_id = 1610612744 AND season = '2021-22'
"""

result_7 = pd.read_sql(query_7, conn)
display(result_7)
```

	avg_points
0	111.0

Query 8: Complete Summary Statistics

Task: Create a summary with COUNT, SUM, AVG, MIN, and MAX for 'pts' for all games in 2021-22.

Hint: Use all 5 aggregate functions in one SELECT statement

In [94]:
```
# TODO: Write your query
query_8 = """
SELECT
COUNT(*) AS games,
SUM(pts) AS total_points,
AVG(pts) AS avg_points,
MIN(pts) AS min_points,
MAX(pts) AS max_points
FROM team_game_stats
WHERE season = '2021-22';
"""

result_8 = pd.read_sql(query_8, conn)
display(result_8)
```

	games	total_points	avg_points	min_points	max_points
0	2460	272115	110.615854	75	158

Query 9: Count Teams by State

Task: How many teams are located in California?

Hint: COUNT(*) from teams WHERE state = 'California'

In [97]:
```
# TODO: Write your query
query_9 = """
SELECT COUNT(*)
FROM teams
WHERE state = "California"
"""

result_9 = pd.read_sql(query_9, conn)
display(result_9)
```

	COUNT(*)
0	4

Query 10: Oldest Team

Task: What is the earliest year_founded in the teams table?

Hint: MIN(year_founded)

In [100…

```
# TODO: Write your query
query_10 = """

SELECT MIN(year_founded) AS oldest_year_founded
FROM teams
"""

result_10 = pd.read_sql(query_10, conn)
display(result_10)
```

	oldest_year_founded
0	1946

Part 2: GROUP BY Queries (8 queries)

Practice grouping data and aggregating by category.

Remember:

- Every non-aggregated column in SELECT must be in GROUP BY
- GROUP BY creates separate groups for aggregation
- Use ORDER BY to sort your results

Query 11: Games Per Team

Task: How many games did each team play in 2021-22?

Hint: SELECT team_id, COUNT(*) ... GROUP BY team_id

In [104…

```
# TODO: Write your query
query_11 = """

SELECT team_id, COUNT(*) AS games_played
FROM team_game_stats
WHERE season = '2021-22'
GROUP BY team_id
ORDER BY games_played DESC
"""
```

```
result_11 = pd.read_sql(query_11, conn)
display(result_11.head(10))  # Show first 10 teams
```

	team_id	games_played
0	1610612766	82
1	1610612765	82
2	1610612764	82
3	1610612763	82
4	1610612762	82
5	1610612761	82
6	1610612760	82
7	1610612759	82
8	1610612758	82
9	1610612757	82

Query 12: Average Points By Team

Task: Calculate average points per game for each team in 2021-22. Sort by highest average first.

Hint: GROUP BY team_id, ORDER BY avg_points DESC

In [107…

```
# TODO: Write your query
query_12 = """

SELECT team_id, ROUND(AVG(pts), 1) AS avg_points
FROM team_game_stats
WHERE season = '2021-22'
GROUP BY team_id
ORDER BY avg_points DESC
"""

result_12 = pd.read_sql(query_12, conn)
display(result_12.head(10))  # Show first 10 teams
```

	team_id	avg_points
0	1610612750	115.9
1	1610612763	115.6
2	1610612749	115.5
3	1610612766	115.3
4	1610612756	114.8
5	1610612737	113.9
6	1610612762	113.6
7	1610612759	113.2
8	1610612751	112.9
9	1610612743	112.7

Query 13: Team Performance with Names

Task: Show team name, games played, and average points for each team in 2021-22.

Hint: JOIN teams with team_game_stats, then GROUP BY

In [110…

```
# TODO: Write your query
query_13 = """

SELECT t.full_name AS team_name,
       COUNT(*) AS games_played,
       ROUND(AVG(g.pts), 1) AS avg_points
FROM team_game_stats g
JOIN teams t ON g.team_id = t.team_id
WHERE g.season = '2021-22'
GROUP BY t.team_id, t.full_name
ORDER BY avg_points DESC
"""

result_13 = pd.read_sql(query_13, conn)
display(result_13.head(10))
```

	team_name	games_played	avg_points
0	Minnesota Timberwolves	82	115.9
1	Memphis Grizzlies	82	115.6
2	Milwaukee Bucks	82	115.5
3	Charlotte Hornets	82	115.3
4	Phoenix Suns	82	114.8
5	Atlanta Hawks	82	113.9
6	Utah Jazz	82	113.6
7	San Antonio Spurs	82	113.2
8	Brooklyn Nets	82	112.9
9	Denver Nuggets	82	112.7

Query 14: Total Points By Team

Task: Calculate total points scored by each team in 2021-22. Include team name.

Hint: SUM(pts), JOIN with teams table, GROUP BY team

In [113…

```
# TODO: Write your query
query_14 = """

SELECT t.full_name AS team_name,
       SUM(g.pts) AS total_points
FROM team_game_stats g
JOIN teams t ON g.team_id = t.team_id
WHERE g.season = '2021-22'
GROUP BY t.team_id, t.full_name
ORDER BY total_points DESC
"""

result_14 = pd.read_sql(query_14, conn)
display(result_14.head(10))
```

	team_name	total_points
0	Minnesota Timberwolves	9507
1	Memphis Grizzlies	9480
2	Milwaukee Bucks	9470
3	Charlotte Hornets	9457
4	Phoenix Suns	9416
5	Atlanta Hawks	9343
6	Utah Jazz	9316
7	San Antonio Spurs	9279
8	Brooklyn Nets	9258
9	Denver Nuggets	9243

Query 15: Season High by Team

Task: Find each team's highest-scoring game in 2021-22. Include team name and sort by highest game.

Hint: MAX(pts), JOIN, GROUP BY, ORDER BY DESC

In [116…

```
# TODO: Write your query
query_15 = """

SELECT t.full_name AS team_name,
       MAX(g.pts) AS season_high_points
FROM team_game_stats g
JOIN teams t ON g.team_id = t.team_id
WHERE g.season = '2021-22'
GROUP BY t.team_id, t.full_name
ORDER BY season_high_points DESC
"""

result_15 = pd.read_sql(query_15, conn)
display(result_15.head(10))
```

	team_name	season_high_points
0	Charlotte Hornets	158
1	San Antonio Spurs	157
2	Los Angeles Clippers	153
3	Washington Wizards	153
4	Memphis Grizzlies	152
5	Brooklyn Nets	150
6	Minnesota Timberwolves	149
7	Houston Rockets	146
8	Los Angeles Lakers	146
9	Boston Celtics	145

Query 16: Win Count by Team

Task: Count how many wins each team had in 2021-22.

Hint: Use SUM(CASE WHEN wl = 'W' THEN 1 ELSE 0 END) to count wins

In [119…

```
# TODO: Write your query
query_16 = """

SELECT t.full_name AS team_name,
       SUM(CASE WHEN g.wl = 'W' THEN 1 ELSE 0 END) AS wins
FROM team_game_stats g
JOIN teams t ON g.team_id = t.team_id
WHERE g.season = '2021-22'
GROUP BY t.team_id, t.full_name
ORDER BY wins DESC
"""

result_16 = pd.read_sql(query_16, conn)
display(result_16.head(10))
```

	team_name	wins
0	Phoenix Suns	64
1	Memphis Grizzlies	56
2	Golden State Warriors	53
3	Miami Heat	53
4	Dallas Mavericks	52
5	Boston Celtics	51
6	Milwaukee Bucks	51
7	Philadelphia 76ers	51
8	Utah Jazz	49
9	Denver Nuggets	48

Query 17: Teams by State

Task: Count how many teams are in each state.

Hint: GROUP BY state from teams table

In [122…

```
# TODO: Write your query
query_17 = """

SELECT state, COUNT(*) AS team_count
FROM teams
GROUP BY state
ORDER BY team_count DESC, state
"""

result_17 = pd.read_sql(query_17, conn)
display(result_17.head(10))
```

	state	team_count
0	California	4
1	Texas	3
2	Florida	2
3	New York	2
4	Arizona	1
5	Colorado	1
6	District of Columbia	1
7	Georgia	1
8	Illinois	1
9	Indiana	1

Query 18: Players Per Team

Task: Count how many player-season records each team has for 2021-22.

Hint: COUNT(*) from player_season_stats, GROUP BY team_id

In [125…

```
# TODO: Write your query
query_18 = """

SELECT team_id, COUNT(*) AS player_season_records
FROM player_season_stats
WHERE season = '2021-22'
GROUP BY team_id
ORDER BY player_season_records DESC
"""

result_18 = pd.read_sql(query_18, conn)
display(result_18.head(10))
```

	team_id	player_season_records
0	1610612760	26
1	1610612765	25
2	1610612757	24
3	1610612764	23
4	1610612763	22
5	1610612756	22
6	1610612739	22
7	1610612738	22
8	1610612753	21
9	1610612752	21